ROTATION: ADOLESCENT ASSESSMENT

Chapter 8 — Adolescent Assessment (12-20 years)

Health promotion of the adolescent and family: growth during puberty, sexual maturation, nutrition needs, body image and eating disorders, injury prevention, immunization and special screens, and psychosocial assessment.

Age range: 12 – 20 years

8.1 Site Profile — Adolescent Health Supervision

The AAP, American Academy of Family Physicians, American Medical Association, and the U.S. Preventive Services Task Force publish similar guidelines for health supervision of adolescents; services must meet physical and emotional needs and are best delivered by providers trained in adolescent care.

BRIGHT FUTURES — ISSUES TO ADDRESS AT EVERY ADOLESCENT HEALTH VISIT

- **Physical growth and development** — physical and dental health, body image, healthy nutrition, physical activity.
- **Social and academic competence** — relationships with peers and family, school performance, interpersonal relationships.
- **Emotional well-being** — mental health and sexuality.
- **Risk reduction** — tobacco, alcohol, other drugs, pregnancy, sexually transmitted infections.
- **Violence and injury prevention**.

CONFIDENTIALITY

The availability of **confidential services** is particularly important for adolescents; most states have provisions for confidential care, and laws vary by state. Establish the age-related policy for your clinical setting before the interview and explain it to the adolescent and family.

8.2 Physical Growth During Puberty

- The **final 20% to 25% of linear growth** is achieved during puberty, and up to **50% of ideal adult body weight** is gained during this time.
- Girls begin puberty and reach maturity about **1½ to 2 years earlier** than boys; the pubertal growth spurt begins as early as 9½ or as late as 14½ years in girls, and as early as 10½ and as late as 16 years in boys.
- The growth spurt reaches its peak at about **12 years in girls** and about **14 years in boys**.
- **Peak height velocity (PHV)** occurs at about 12 years of age in girls, around 6 to 12 months **before menarche**; PHV is used as a predictor of menarche, and height at menarche predicts ultimate adult height. Few girls grow more than 5 cm (2 inches) after menarche; growth in girls' height usually ceases 2 to 2½ years after menarche.
- **Boys** typically reach PHV at about 14 years, after growth of the testicles and penis and appearance of axillary and mature pubic hair; growth in height ceases at 18 or 20 years.
- **Peak weight velocity** occurs about 6 months after PHV in girls (and about 6 months after menarche); in boys the weight and height spurts occur simultaneously.
- Leg-length increase precedes trunk growth by about 6 to 9 months and shoulders/chest by about 1 year: "outgrow **shoes** first, then **pants**, then **shirts**".
- **Body composition:** girls' fat mass increases markedly in early puberty and continues to increase after menarche; boys have a peak deceleration in fat-mass accumulation at the growth spurt, with significantly higher lean body mass overall.
- On average, girls gain 5-20 cm (2-8 inches) in height and 7-25 kg (15.5-55 lb) in weight during adolescence. In the U.S., onset of Tanner stage 2 genital development in boys (mean): non-Hispanic white 10.14 yr, African-American 9.14 yr, Hispanic 10.04 yr.

Figure 8.1 places the pubertal events described above on a single timeline: peak height velocity at about 12 years in girls and 14 years in boys, the menarche band, and the growth-spurt details for each sex.

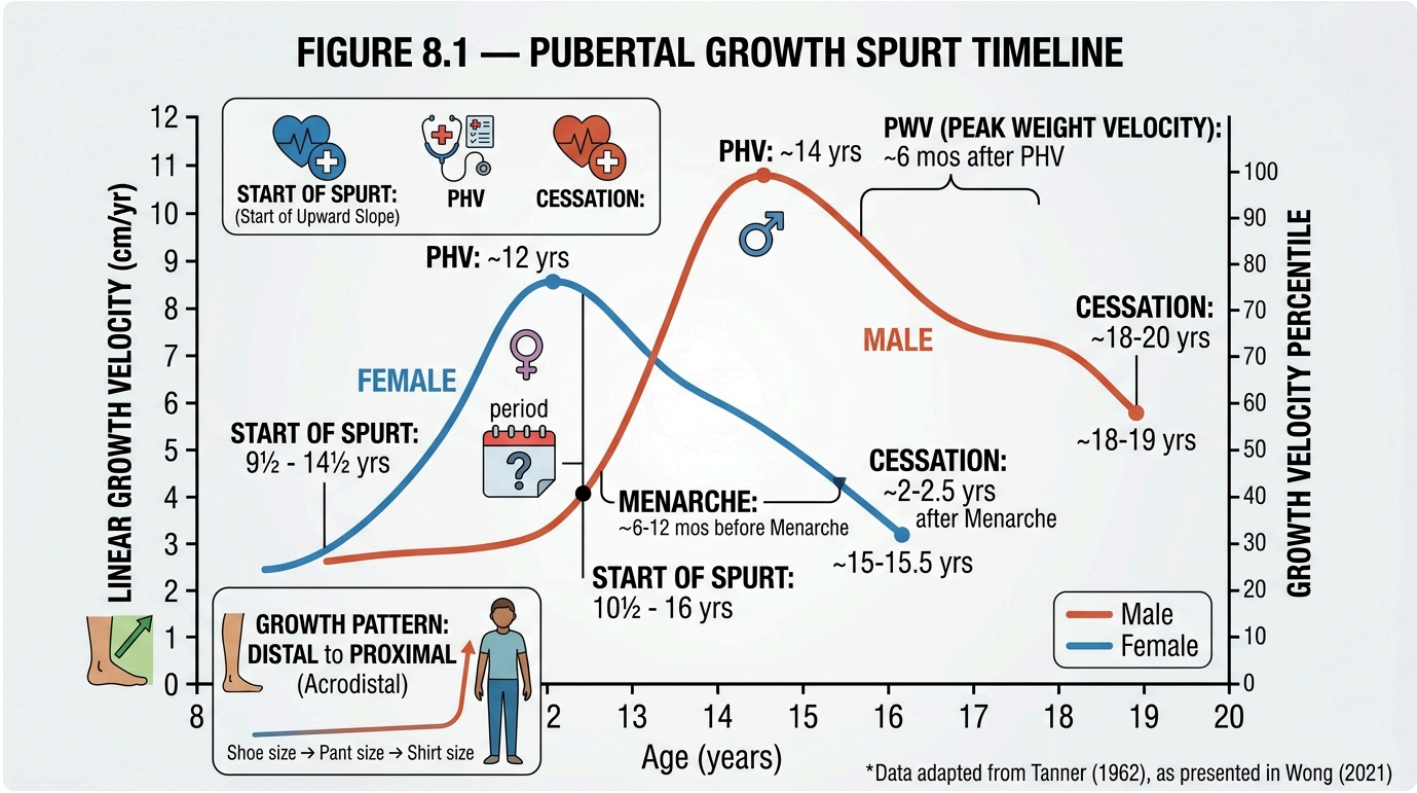


Figure 8.1 Timeline of the pubertal growth spurt in girls and boys.

8.3 Sexual Maturation — Tanner Staging

- Tanner stages describe pubertal development in numbered stages; in females they describe development based on **breast size and the shape and distribution of pubic hair**, in males on **genital development and pubic hair**; stage 1 = prepubertal, stage 5 = adult.
- Example entries (boys): *Stage 1 (prepubertal):* no pubic hair — essentially the same as during childhood, no distinction between hair on the pubis and over the abdomen. *Stage 4:* increased size of the penis with growth in diameter and development of the glans; glans larger and broader; scrotum darker; pubic hair more abundant with curling.
- **Fertility:** from a clinical perspective an adolescent should be considered potentially fertile with a first menstrual period or a first ejaculation; viable sperm production tends to follow the first ejaculation by about 1 year.
- Boys do not experience a discrete event analogous to menstruation; the capacity to ejaculate appears approximately 1 year after initial testicular enlargement and the appearance of pubic hair.

The full breast-and-pubic-hair staging tables are not reproduced here; consult your course's printed staging charts for the complete figures while practising the examination.

8.4 Nutrition in Adolescence

Point	Practice implication
Puberty can **double** nutritional requirements for **iron, calcium, zinc, and protein**	Assess intake at every visit; high-need foods first (dairy, protein, whole grains, fruits, vegetables)
Girls are particularly susceptible to **iron deficiency at menarche**	Screen dietary iron; anemia signs; encourage iron-rich foods with vitamin C
Maximum bone mass is acquired during adolescence; calcium deposited now determines later osteoporosis risk	Encourage calcium-rich intake; discourage dietary substitutes that displace dairy
Common pattern: excess calories, sugar, fat, cholesterol, sodium; inadequate folic acid, vitamin B6, vitamin A, iron, calcium, zinc — especially girls and low-SES teens	Teach balanced food choices (as in the school-age nutrition section); watch fast-food and soft-drink displacement
Weight-conscious behavior and **overweight** have increased among adolescents	Plot growth on CDC charts; discuss weight status without body shaming

8.5 Body Image & Eating Disorders

- Body image is important for adolescent girls; a common behaviour is **extensive dieting at a time when physical changes are occurring**, and a focus on slimness and dieting may **trigger the development of eating disorders**.
- **Anorexia nervosa and bulimia nervosa** occur among adolescents; screen with non-judgmental questions about meal patterns, dieting behaviour, and weight concerns.
- Body image and gender-role definition are **nearly secured** by late adolescence; identity consolidates.
- Acne and feelings of sadness or depression are among the health concerns adolescents report.
- This guide covers the detection and referral signals for eating disorders; diagnosis and treatment follow specialist protocols.

8.6 Injury Prevention & Safety

- **Injuries kill more U.S. adolescents than any other single cause**; unintentional injury accounted for **48% of deaths** among 12-19-year-olds (1996-2005), and its share rose with age.
- **Motor-vehicle crashes** are the single greatest source of unintentional injury and death in young people; contributory factors: lacking driving experience and maturity, following too closely, speeding, other teen passengers in the car, alcohol use.
- **Homicide** is the second leading cause of death among all U.S. adolescents and the most likely cause for African-American teenagers; in 2007, **82% of homicides** among 13-19-year-olds were firearm-related.
- In 2011, 5.1% of adolescents nationwide reported carrying a weapon on one or more days in the past month (down from 27% in 1991); teens targeted for repeated teasing/harassment are more likely to skip school, report depression symptoms, attempt suicide, and **bring weapons to school**.
- **Sports:** participation physical examinations at least every 2 years; teach and model helmet use for cycling, skating, and helmeted sports (a helmet reduces head injury ~45% and brain injury ~33%).
- Cover injury-prevention counseling at every visit per the Bright Futures "violence and injury prevention" item.

8.7 Adolescent Immunizations & Special Screens

Vaccine / screen	Recommendation
Meningococcal (MenACWY)	Adolescents 11-12 years with a booster dose before age 16; if not previously vaccinated, one dose between 13 and 18 years
HPV	9-valent vaccine (Gardasil 9) for girls and boys; routine at 11-12 years with a minimum age of 9; two doses if the series starts at 9-14 years, three doses if it starts at 15 years or later or in immunocompromised persons
Hepatitis B	Catch-up for any adolescent who has not received three doses
Hepatitis A	Targeted — adolescents living where programs target older children, at increased risk, or when immunity is desired
Influenza	Annual vaccination (live attenuated or inactivated) for all children and adolescents

Varicella	Assess history; vaccinate those with no previous infection/history — two doses ≥4 weeks apart for adolescents 13 years or older
Tuberculin skin test	If exposed to active TB, lived in a homeless shelter, incarcerated, from/came from a high-prevalence TB area, or currently work in a health care setting
Blood pressure	BP rises from the onset of adolescence until the end of pubertal growth, especially in males; ~1% of adolescents have sustained hypertension — measure annually and use the appropriate-size cuff

The adolescent schedule changes from time to time — verify the ages above against the current official printed schedule when you chart. The scoliosis and vision/hearing methods from Chapters 4 and 7 continue to apply in the adolescent clinic.

8.8 Psychosocial Assessment & Health Education

INTERVIEW DOMAINS

- **School:** recent grades, absences, suspensions, repeated grades; school-related strengths and deficits; involvement in activities.
- **Peers:** quality of peer relationships — may identify socially isolated teens.
- **Future plans:** education/employment and career choices — a window into sources of stress.
- **Strengths:** teach coping-with-stress skills and involvement in personally meaningful activities to support emotional health.
- **Emotional well-being:** sadness, depression, suicidal ideation; adolescents whose personal fable stresses their uniqueness are at higher risk for depression and suicidal ideation, while a strong sense of invulnerability predicts risky behaviour.
- **Risk behaviours:** below-average grades correlate with tobacco/alcohol use, unprotected intercourse, and suicide attempts; violence and harassment on the basis of race, gender, or sexual orientation affect more than half of students.

These domains are commonly memorised with the HEEADSSS mnemonic used in adolescent teaching materials — Home, Education, Eating, Activities, Drugs, Sexuality, Suicide-depression, Safety — and every question it prompts maps to the content here.

Chapter 8 Learning Check

Self-assessment questions (answer in your notebook, then open)

End of Chapter 8.